

TITLE PAGE

- **Problem Statement ID –**
- **Problem Statement Title-**
- **Theme-**
- **PS Category- Software/Hardware**
- **Team ID-**
- **Team Name (Registered on portal)**



IDEA TITLE

• Proposed Solution (Describe your Idea/Solution/Prototype)

1. The Idea:

Solving Medical Fragmentation

- Women's health data is scattered.
- Records are stored across doctors, PDFs, and prescriptions.
- No single view shows the complete health history.
- No clear timeline connects symptoms and treatments.

→ Longitudinal AI Health Memory

2. The Solutionheading:

AI Intelligence Layer

- Report Extraction: Converts PDFs and scans into structured data.
- Health Timeline: Connects symptoms, tests, and treatments.
- Pattern Detection: Finds trends, gaps, and possible connections.

→ Personalized Health Summary

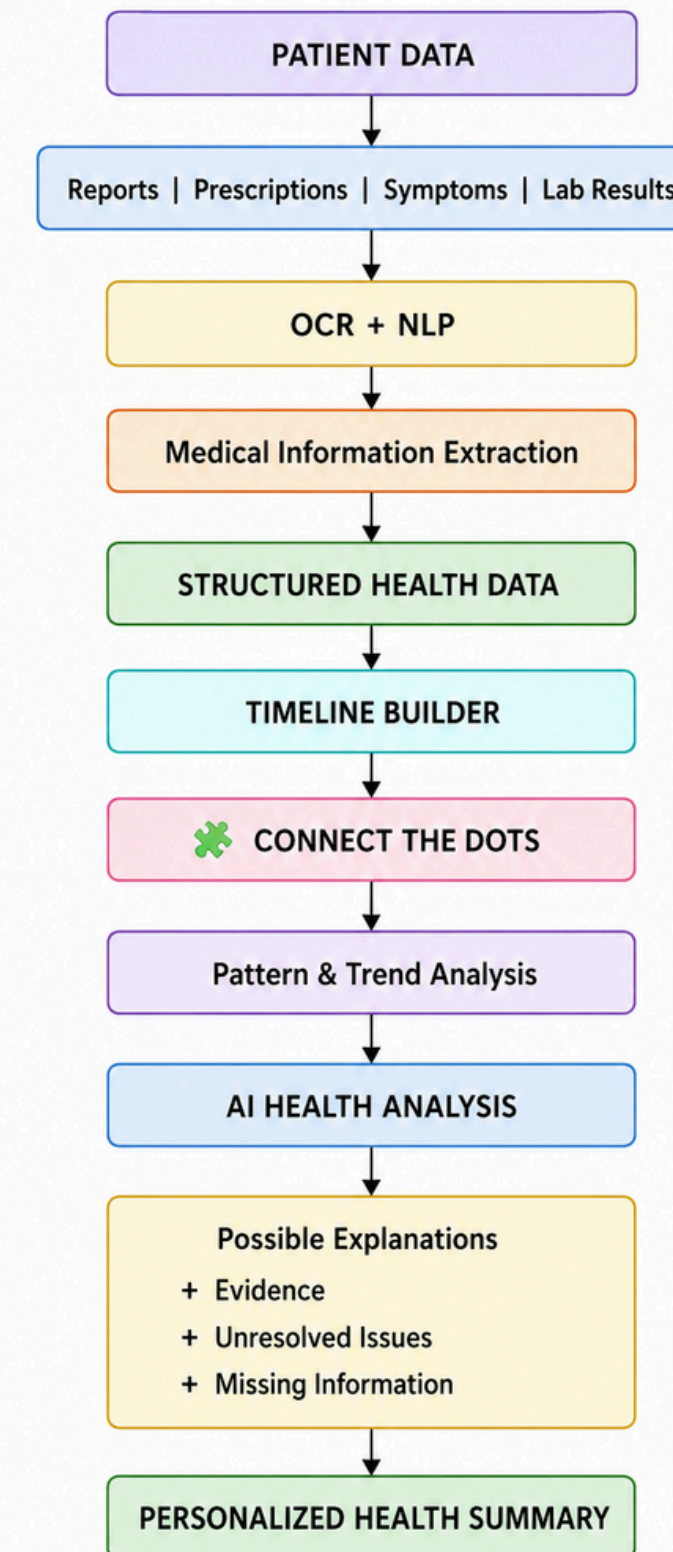
3. The Prototype

MVP Features

- Document Parser: Upload reports and extract data.
- Timeline UI: View health history and symptom trends.
- Pattern & Gap Detector: Highlights trends and missing information.

→ AI Pattern & Gap Engine

- Frontend: React / Next.js
- Backend: Python + FastAPI
- AI/ML: LLM + NLP
- Document Processing: OCR
- Knowledge Retrieval: RAG + Vector Database
- Knowledge Graph: Neo4j
- Database: PostgreSQL
- Visualization: Recharts / Chart.js



Feasibility

- **Software-Based:** No special hardware is required.
- **Proven Technology:** Uses existing AI, OCR, and NLP tools.
- **Cloud-Based:** Easy to access, scale, and deploy.
- **Focused Data:** Uses curated women's health datasets.
- **Safe Testing:** Uses synthetic medical records for development.

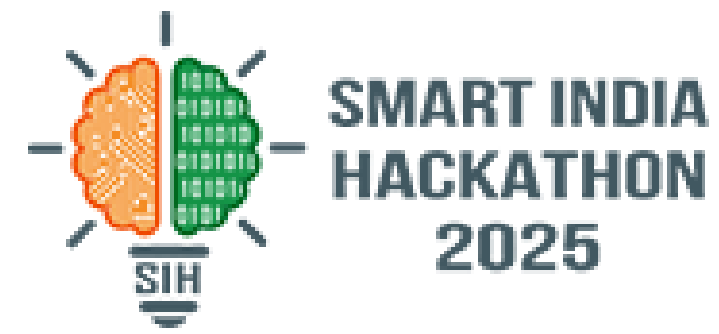
Key Challenges

- **Medical Accuracy:** AI may give wrong or misleading medical conclusions.
- **Document Quality:** Scanned reports may be blurry or difficult to read.
- **Missing Information:** Patient history or important health details may be incomplete.
- **Data Privacy:** Medical records contain sensitive personal health information and need strong protection.

Our Approach

- **Evidence-Based AI:** Gives reliable answers with confidence scores.
- **User Verification:** Users verify all extracted medical data.
- **Missing Data Alerts:** Clearly highlights missing health information.
- **Strong Security:** Uses encryption and secure login.
- **Safe Insights:** Explains health patterns without giving unsupported diagnoses.

IMPACT AND BENEFITS



- Complete Health History in One Place
- Understand Your Health Journey
- Identify Recurring Symptoms
- Build Long-Term Health Awareness
- Track Treatment Responses
- Connect Related Health Events
- Never Lose Important Medical Information
- Track Changes Over Time

- Details / Links of the reference and research work

World Health Organization (WHO): Frameworks addressing gender bias in medical data, fragmentation of records, and standards-based digital health guidelines.

WHO Women's Health & Life-Course Care

WHO SMART Digital Health Guidelines

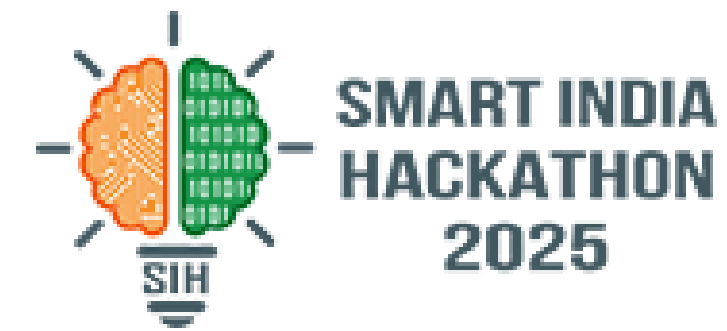
World Health Organization, "International Statistical Classification of Diseases and Related Health Problems (11th ed.)," WHO.int, 2022. [Online]. Available:

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World Health Organization, "Global Strategy on Digital Health 2020–2025," World Health Organization, Geneva, Switzerland, 2021. [Online]. Available:

<https://www.who.int/teams/digital-health-and-innovation/smart-guidelines>

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.